10 (a) Stars more massive than the Sun can eventually form black holes.

Describe how a black hole can be formed from a more massive star.

.....

.....

.....

.....

..... [3]

(b) The star system V404 Cygni contains a black hole. The system is approximately 7800 light-years from the Earth.

(i) Describe what is meant by a light-year.

.....

..... [1]

(ii) Calculate the approximate distance from V404 Cygni to the Earth in km.

distance = ... km [2]

[Total: 6]

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